

desired

T_{targets}
 $p_{\text{CoM},d}$
 $q_{\text{posture},d}$

SE(3) / CoM tasks

$\xi_e = \text{Log}(TT_d^{-1})^\vee$
torso · CoM · posture

controller

whole-body QP

$\min_{\ddot{q}, \tau, \lambda, s} \mathcal{J}$
dynamics · contact ·
friction/CoP · limits

τ

Unitree G1

MuJoCo plant
floating-base · fixed-foot DS

F_{push}

state, contacts
measured GRF

$q, \dot{q}$, contacts

Evaluation

recovery · slip · basin